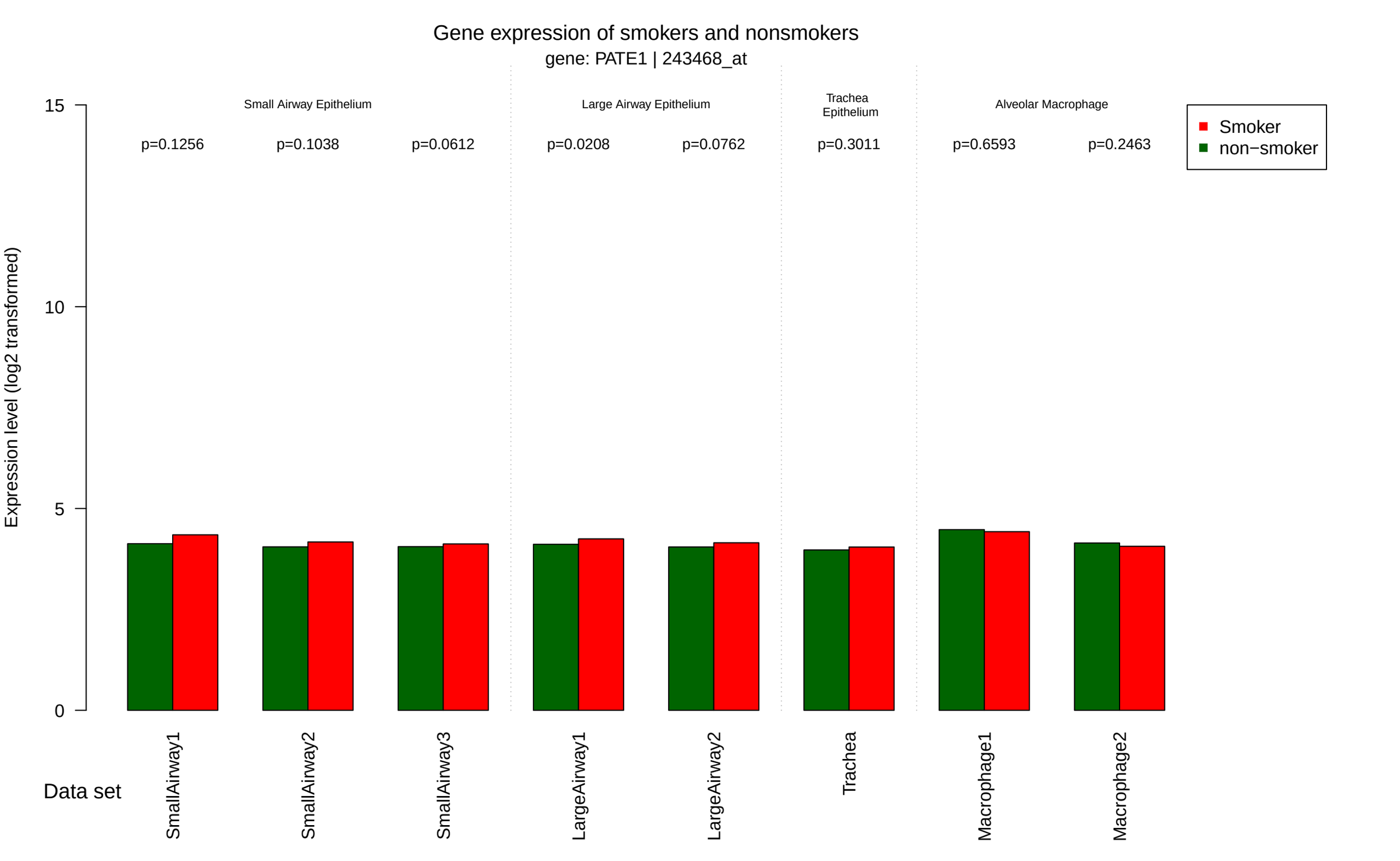
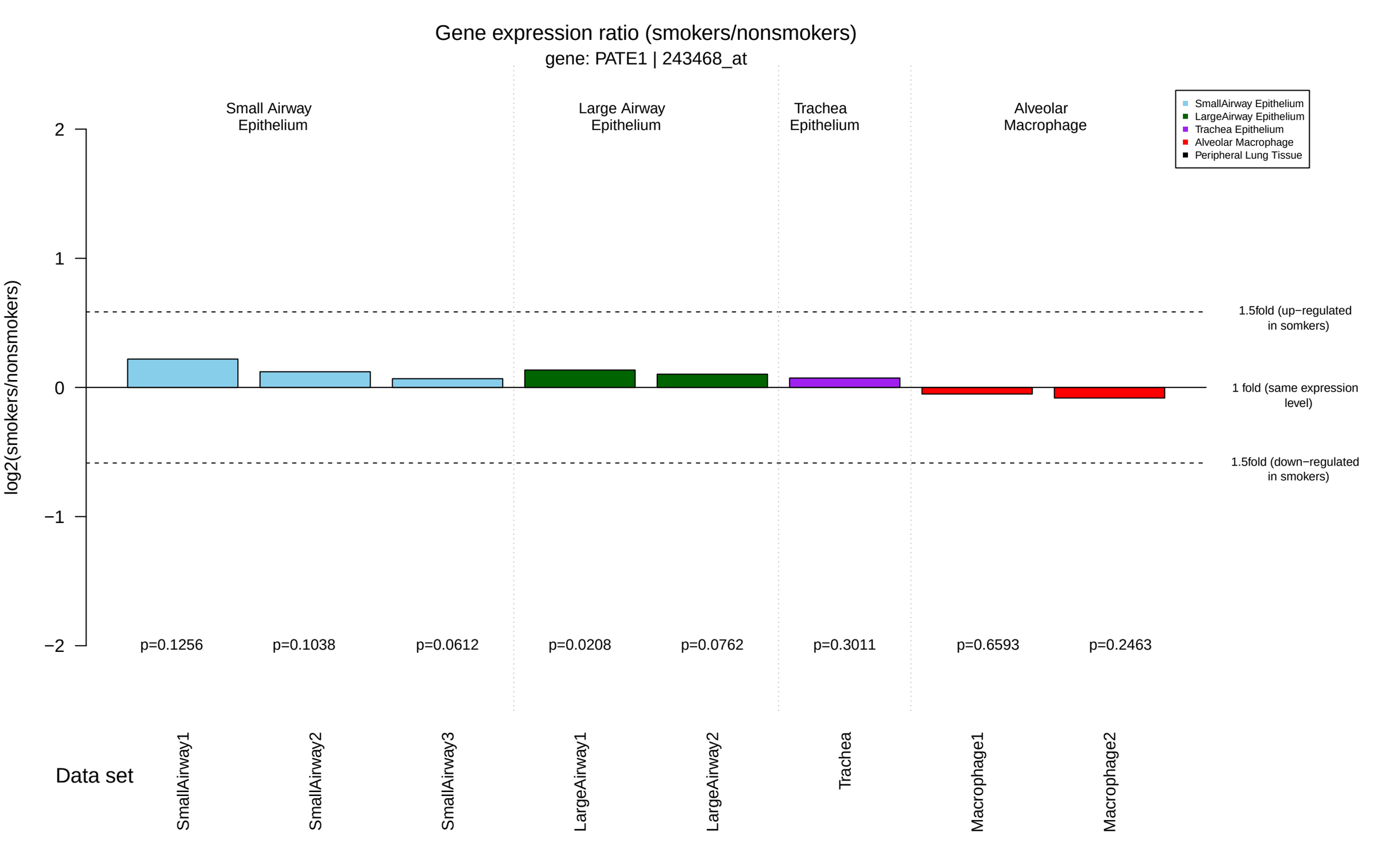
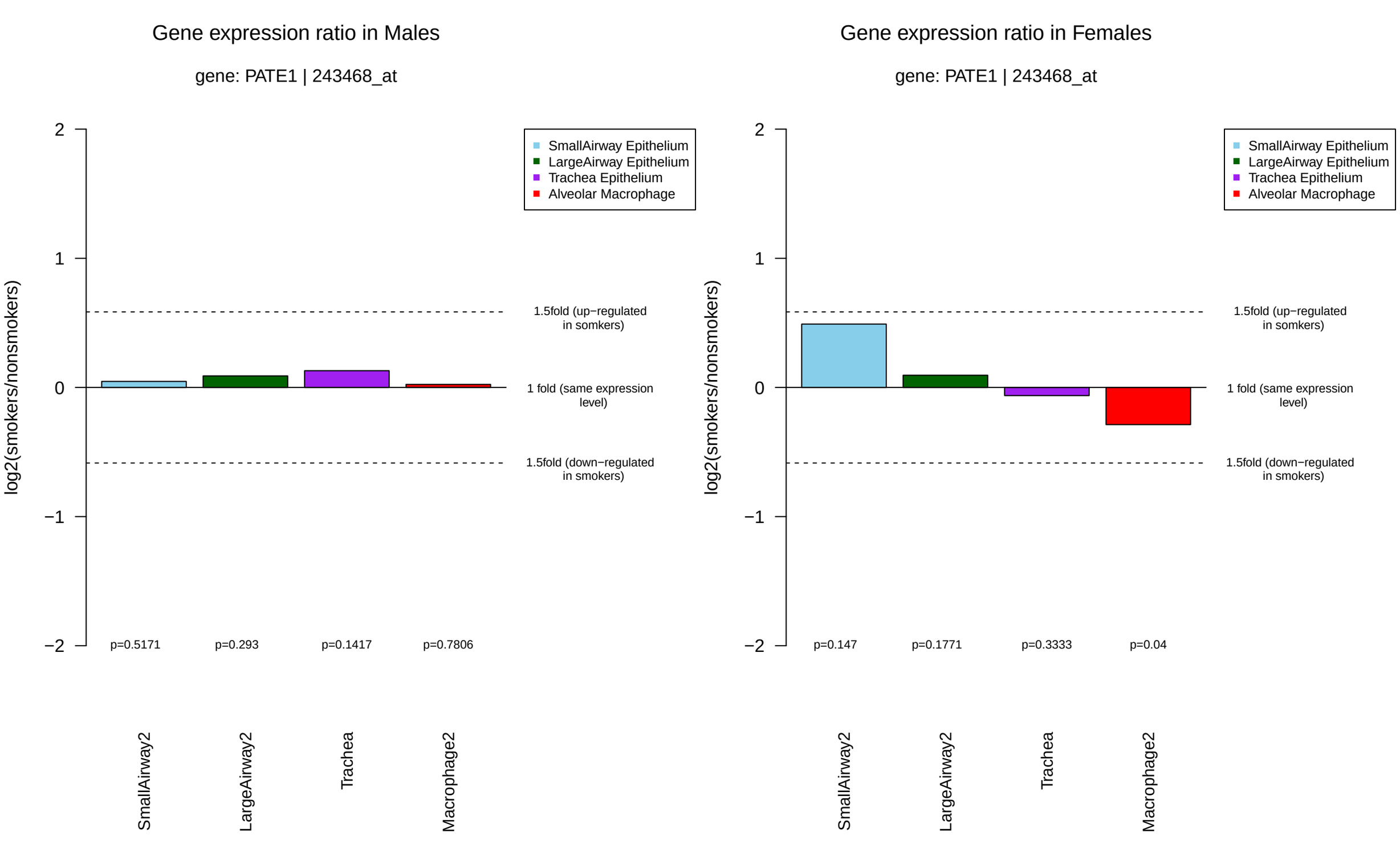
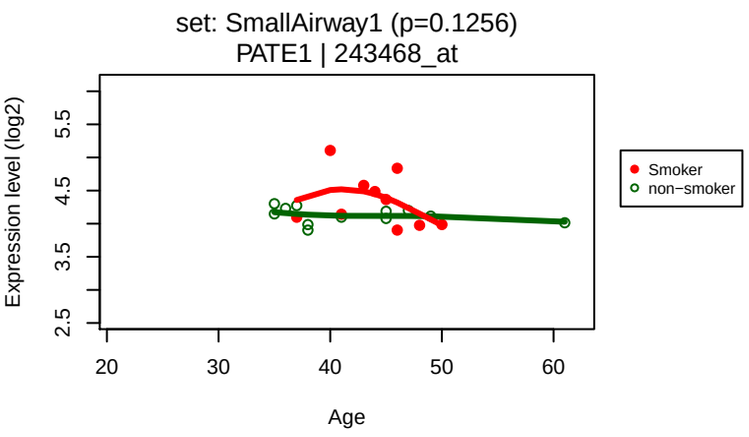
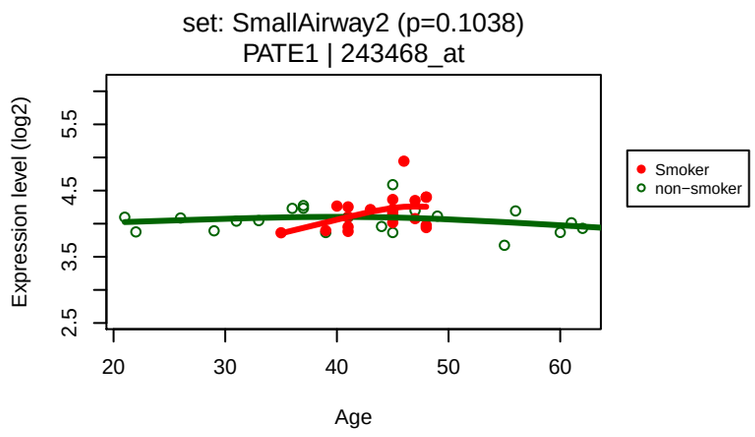
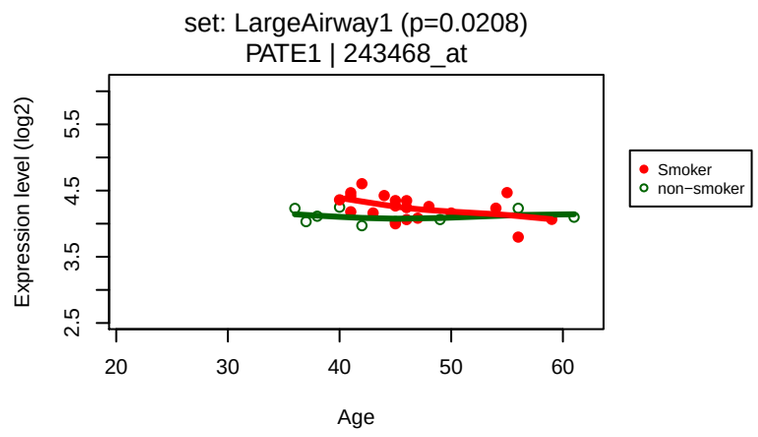

Gene expression of smokers and nonsmokers
gene: PATE1 | 243468_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.1256
p=0.1038
p=0.0612
p=0.0208
p=0.0762
p=0.3011
p=0.6593
p=0.2463
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

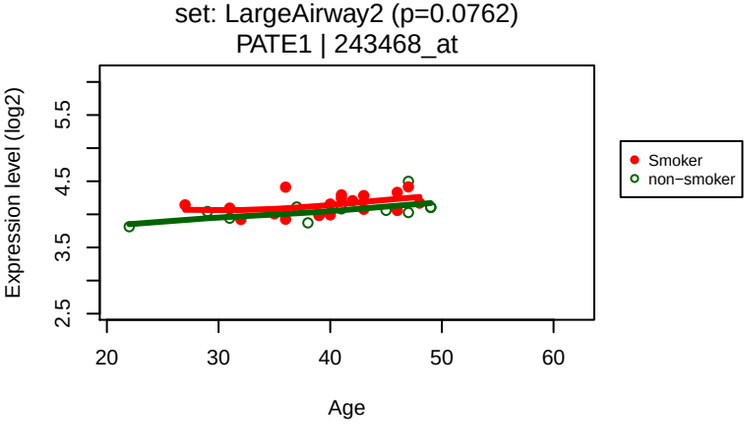

Gene expression ratio (smokers/nonsmokers)
gene: PATE1 | 243468_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.1256
p=0.1038
p=0.0612
p=0.0208
p=0.0762
p=0.3011
p=0.6593
p=0.2463
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

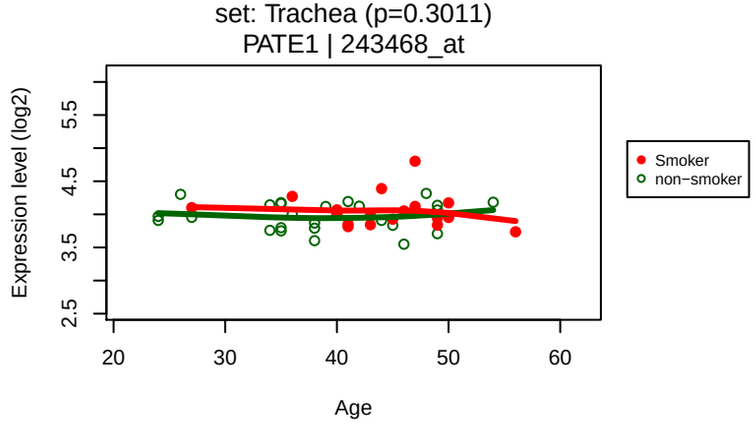

Gene expression ratio in Males
gene: PATE1 | 243468_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.5171
p=0.293
p=0.1417
p=0.7806
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: PATE1 | 243468_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.147
p=0.1771
p=0.3333
p=0.04
SmallAirway2
LargeAirway2
Trachea
Macrophage2

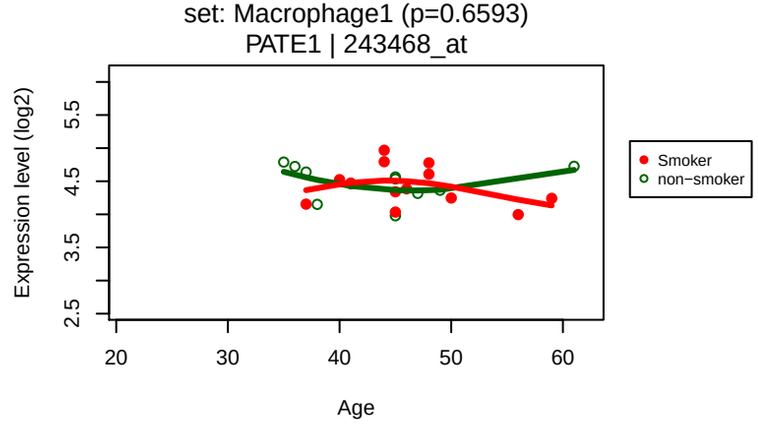

set: SmallAirway1 (p=0.1256)
PATE1 | 243468_at
Expression level (log2)
2.5
3.5
4.5
5.5
20
30
40
50
60
Age
Smoker
non−smoker

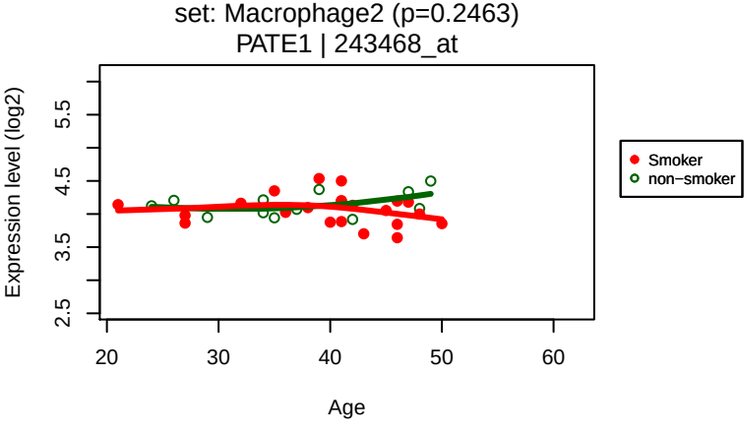

set: SmallAirway2 (p=0.1038)
PATE1 | 243468_at
Expression level (log2)
2.5
3.5
4.5
5.5
20
30
40
50
60
Age
Smoker
non−smoker

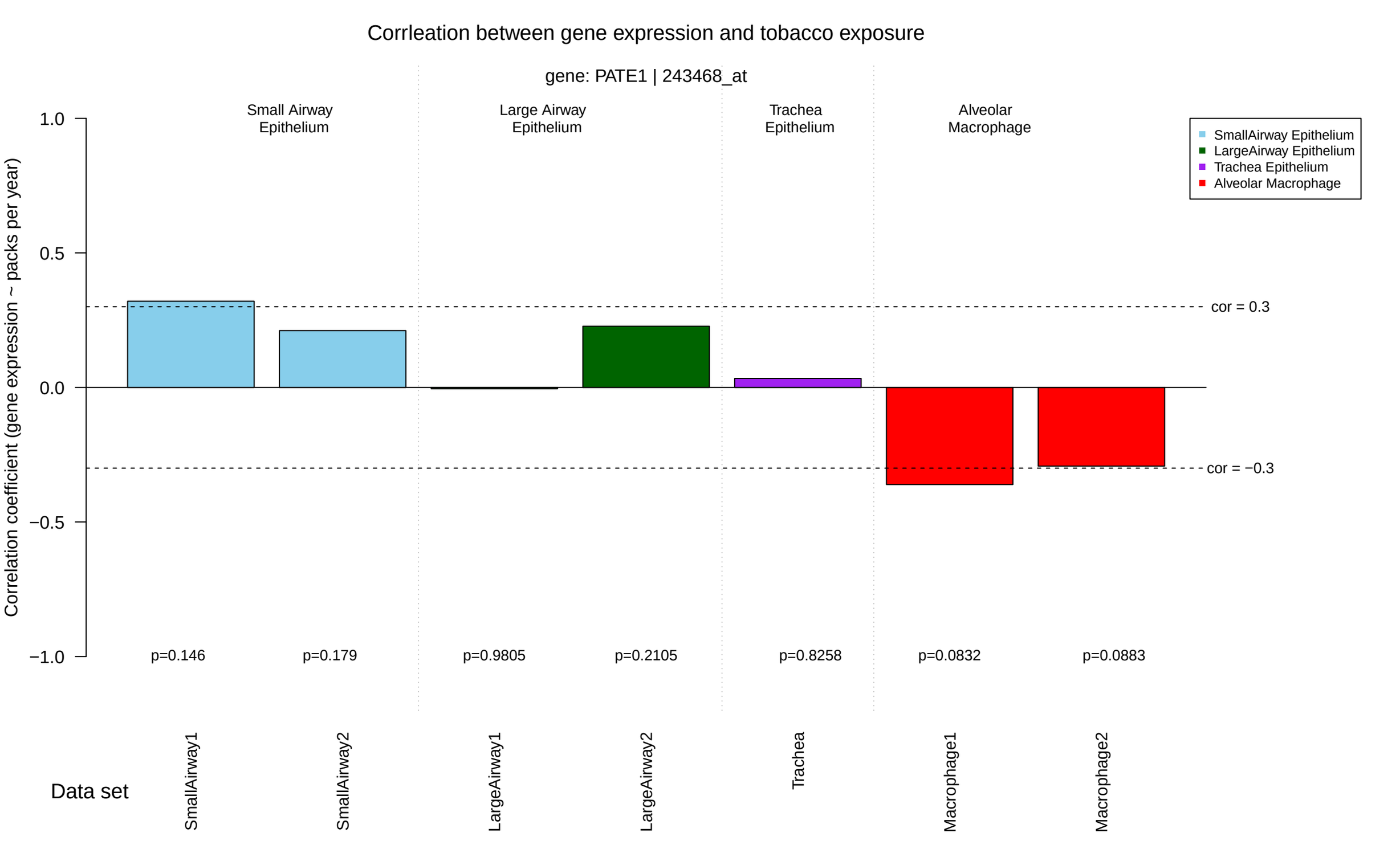

set: LargeAirway1 (p=0.0208)
PATE1 | 243468_at
Expression level (log2)
2.5
3.5
4.5
5.5
20
30
40
50
60
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.0762)
PATE1 | 243468_at
Expression level (log2)
2.5
3.5
4.5
5.5
20
30
40
50
60
Age
Smoker
non−smoker


set: Trachea (p=0.3011)
PATE1 | 243468_at
Expression level (log2)
2.5
3.5
4.5
5.5
20
30
40
50
60
Age
Smoker
non−smoker


set: Macrophage1 (p=0.6593)
PATE1 | 243468_at
Expression level (log2)
2.5
3.5
4.5
5.5
20
30
40
50
60
Age
Smoker
non−smoker


set: Macrophage2 (p=0.2463)
PATE1 | 243468_at
Expression level (log2)
2.5
3.5
4.5
5.5
20
30
40
50
60
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: PATE1 | 243468_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
Correlation coefficient (gene expression ~ packs per year)
p=0.146
p=0.179
p=0.9805
p=0.2105
p=0.8258
p=0.0832
p=0.0883
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2